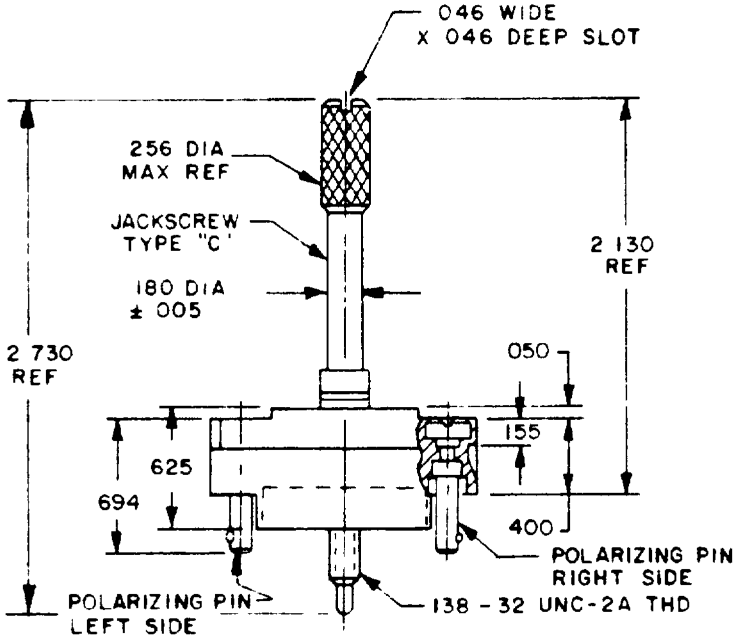
MIL-C-28731/27B
4 June 1985
SUPERSEDING
MIL-C-28731/27A
25 May 1984

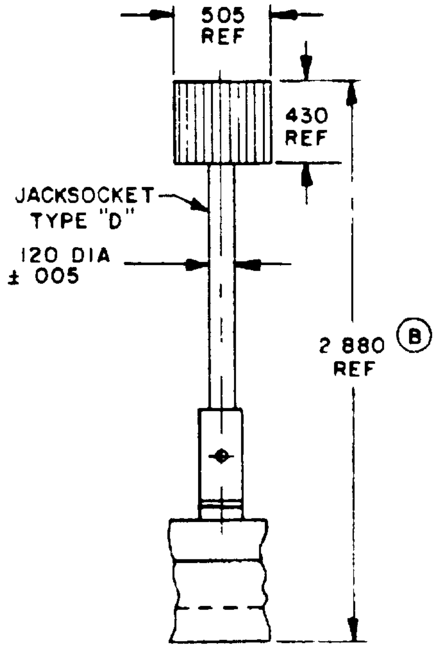
MILITARY SPECIFICATION SHEET

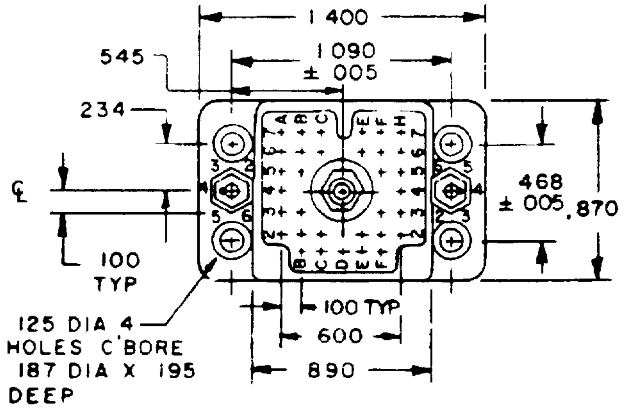
CONNECTORS, ELECTRICAL, INSERT (INSULATOR), MALE RECTANGULAR, POLARIZED, CENTER SCREW LOCK, FOR 33 REMOVABLE TYPE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

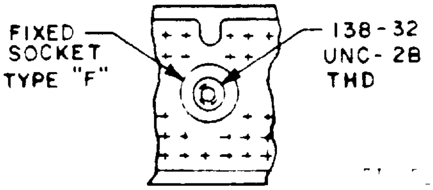
The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.


046 WIDE
X 046 DEEP SLOT
256 DIA
MAX REF
JACKSCREW
TYPE "C"
180 DIA
± 005
2 130
REF
2 730
REF
050
155
625
694
400
POLARIZING PIN
RIGHT SIDE
138 - 32 UNC-2A THD
POLARIZING PIN
LEFT SIDE


505
REF
430
REF
JACKSOCKET
TYPE "D"
120 DIA
± 005
2 880
REF
Ⓑ


1 400
545
1 090
± 005
234
℄
100
TYP
468
± 005
.870
125 DIA 4
HOLES C'BORE
187 DIA X 195
DEEP
100 TYP
600
890

INCHES	MM	INCHES	MM
.005	0.13	.430	10.92
.046	1.17	.468	11.89
.050	1.27	.505	12.83
.080	2.03	.545	13.84
.100	2.54	.600	15.24
.120	3.05	.625	15.88
.125	3.18	.694	17.62
.155	3.94	.870	22.10
.180	4.57	.890	22.61
.187	4.75	.940	23.88
.195	4.95	1.090	27.69
.234	5.94	1.400	35.56
.256	6.50	2.130	54.10
.400	10.16	2.710	68.83
		2.730	69.34


FIXED
SOCKET
TYPE "F"
138 - 32
UNC - 2B
THD

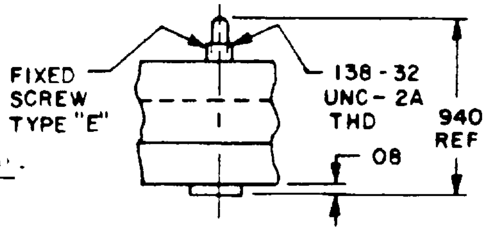
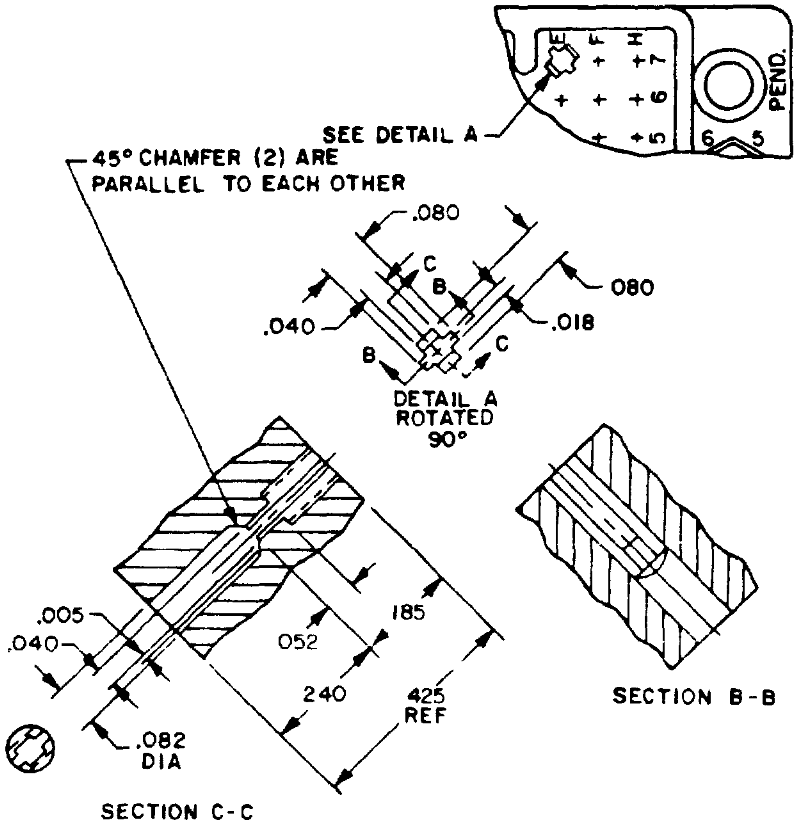

FIXED
SCREW
TYPE "E"
138 - 32
UNC - 2A
THD
940
REF
08

FIGURE 1. [illegible]

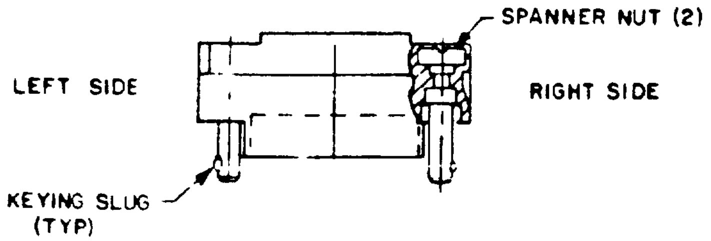
◯ denotes changes

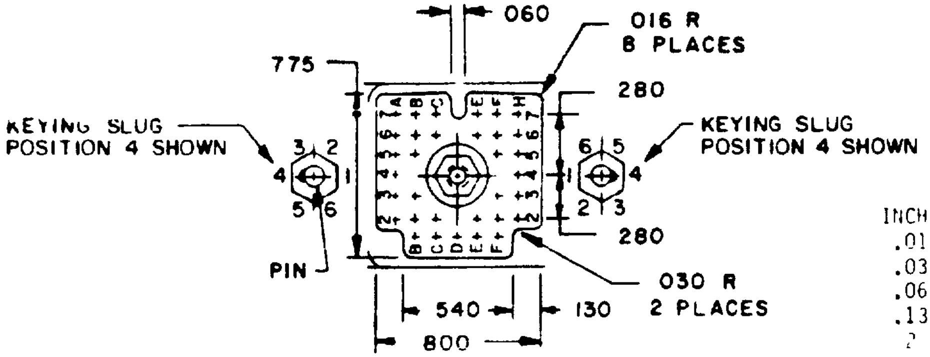

SEE DETAIL A
E
F
H
7
6
5
PEND.
6
5
45° CHAMFER (2) ARE PARALLEL TO EACH OTHER
.080
C
B
080
.040
.018
B
C
DETAIL A ROTATED 90°
.005
.040
185
052
240
425 REF
.082 DIA
SECTION C-C
SECTION B-B

[illegible]

INCHES	MM
.005	.13
.018	.46
.040	1.02
.052	1.32
.080	2.03
.082	2.08
.185	4.70
.240	[illegible]
[illegible]	[illegible]

FIGURE [illegible]. [illegible] - Continued.


SPANNER NUT (2)
LEFT SIDE
RIGHT SIDE
KEYING SLUG
(TYP)


060
016 R
8 PLACES
775
280
KEYING SLUG
POSITION 4 SHOWN
KEYING SLUG
POSITION 4 SHOWN
3 2
4 1
5 6
6 5
1 4
2 3
A B C E H
1 2 3 4 5 6 7
B C D E F
PIN
280
030 R
2 PLACES
540
130
800

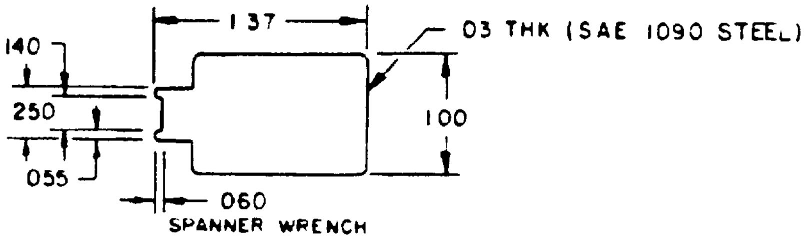
INCHES	MM
.016	0.41
.030	0.76
.060	1.52
.130	3.30
.280	7.11
.540	13.72
.775	19.68
.800	20.32

POLARIZING KEY MATING CODE	
P (MALE) Position	P (FEMALE) Position
4	1
	2
	3
	4
	5
	6

POLARIZING POSITION CHANGE

Keying position on both polarizing pins can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting pin in one of six positions then retightening spanner nut (clockwise).

NOTE Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below


1 37
03 THK (SAE 1090 STEEL)
140
250
100
055
060

SPANNER WRENCH

INCHES	MM
.03	.76
.05	1.27
.060	1.52
.140	3.56
.250	6.35
1.00	25.4
1.37	34.8

CONFIGURATION OPTIONAL
DIMENSIONS REF ONLY

Configuration B - Connector [illegible] (layout)

FIGURE 1. Connector, male rectangular - Continued

NOTES

1 Dimensions are in inches.

2 Metric equivalents are given for general information only.

3 Tolerances between any two contact centers shall be ± .004 (0.10 mm).

4 Flatness, squareness, and parallelism shall be within dimensional tolerances.

5 Contact hole details shall be in accordance with configuration B.

6 Mating connector information.
M28731/27-0001 mate with M28731/28-0003 and -0005
M28731/27-0002 mate with M28731/28-0001 and -0006
M28731/27-0003 and -0005 mate with M28731/28-0001
M28731/27-0004 and -0006 mate with M28731/28-0002.

7 Polarization information.
Polarization pins are provided for indexing to preclude intermixing of adjacent connectors. By using a standard type spanner wrench, polarization pins can be rotated to desired position. For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide (configuration A (see marking, note 13d). For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment.

8 Contact and terminal designations.
Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1.

9. Tolerance shall be ± .010 (0.25 mm) unless otherwise specified.

10. Patent notice.
Patent No. 3,675,195 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.

11 For direct government acquisition crimp contacts M28731/35-0001 are furnished in a plastic bag in each unit package (see marking, note 13a).

12 For direct government acquisition wrappost contacts are M28731/36-0001 and are factory installed in connector body (see marking, note 13a).

13 Marking.
a Connectors supplied directly to the government shall be marked with the military part number (see table I). Example: M28731/27-0001.
b Connectors supplied with contacts to government end users shall be marked with the military part number (see table I). Example: M28731/27-0001.
c Connectors supplied without contacts to government end users shall be marked as follows: Example: M28731/27D (screw or socket type from table I). The user shall complete the remaining portion of the military part number as shown in table I based upon contact type he assembles. Example: M28731/27D-0002.
d Polarization position marking is not required unless otherwise specified. When required polarization marking shall be in accordance with the connector polarization guide (see configuration B). The marking shall appear following the part number.

14 Applicable contact insertion and removal tools.

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Insertion	MIL-[illegible]
Removal	MIL-[illegible]

15. Recommended hardware (not furnished) for panel mounting: (No. 4) .112-40-UNC-2A (length to suit) bolt with No. 4 washer and mating nut.

FIGURE 1. Connector, male, rectangular - Continued.

REQUIREMENTS

Dimensions and configuration See figure 1

Mating and unmating forces (torque) 48 inch-ounces, maximum.

Dielectric withstanding voltage 1,000 volts rms (sea level)

Mechanical shock (specified pulse): Applicable.

Contact retention 6 pounds

Contact separation force 1 ounce, minimum.

Contact current rating 5 amperes, 22 AWG wire.

Part number see table I.

TABLE I Part number information.

Military part number	Actuating hardware	Polarizing pin position		Contact type
		Left side	Right side	
M28731/27-0001	Jackscrew Type C	4	4	Crimp, MIL-C-28731/35 1/
M28731/27-0002	Jack socket Type D	4	4	Crimp, MIL-C-28731/35 1/
M28731/27-0003	Fixed socket Type F	4	4	Crimp, MIL-C-28731/35 1/
M28731/27-0004	Fixed screw Type E	4	4	Crimp, MIL-C-28731/35 1/
M28731/27-0005	Fixed socket Type F	4	4	Wrappost, MIL-C-28731/36 2/
M28731/27-0006	Fixed screw Type E	4	4	Wrappost, MIL-C-28731/36 2/

1/ See figure 1, note 11
2/ See figure 1, note 12

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AL, AV, MI
Navy - MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity
Navy - EC

(Project 5935-3198-10)